THESIS · EXPERIMENT DESIGN FOR APPROVAL

How much welfare do imperfect price forecasts cost?

Equivalent storage bids built from a noisy forecast, compared with Baldick's SOC-based benchmark

1 **Bid families**

from perfect information to none

2 **Error model**

size, persistence, bias, common share

3 **Pipeline and metrics**

pilot results and decisions

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What theory already answers, and what only experiments can

THE QUESTION

Proven (equivalence note)

- With the true prices $\lambda^\star$, every battery can build bids that reproduce Baldick's prices and dispatch, also with many batteries (Theorem 1).
- If no SOC limit ever binds, technical data alone are enough: no forecast needed (Corollary 2).
- Prices fix the bids, but when the battery sets the price, its quantity must also be right (Remark 1).

Open (experiments)

- How much welfare is lost when bids are built from a forecast $\hat{\lambda}$ instead of $\lambda^\star$?
- Which errors hurt most: size, persistence over hours, bias, or errors shared by all batteries?
- When is a forecast worth more than no forecast at all (technical data only)?
- How much of the loss is price information and how much quantity information?

Why it matters The welfare gap is the value of the SOC-based reform: if it is small, well-built interval bids are enough.

Four ways a battery can bid in the current design

1 · BID FAMILIES

Family	What the battery knows	How its bids are built	Role
B★	true prices λ★ and Baldick's quantities	Φ(θ, λ★), plan = Baldick's schedule	upper bound: ΔSW = 0 (sanity check)
B★p	true prices λ★	Φ(θ, λ★), plan = its own self-schedule	isolates the quantity-information loss
B fcst	**a forecast λ̂ (with error)**	**Φ(θ, λ̂), pre-limit rule, own plan**	**main object of the experiments**
B priv	technical data only	value of energy without SOC limits, full capacity	lower bound: "no forecast"

Technical data only (Corollary 2, no rents):

$$u_t = \alpha\,\gamma^{T-t} w - \epsilon, \qquad c_t = \zeta + \beta\,\gamma^{T-t} w$$

Quantity rule: the planned quantity is offered just inside the bid, the rest just outside (δ = 0.01 $/MWh), so ties are broken in favour of the plan:

$$\left(u_t + \delta,\ p_t^{\circ}\right),\ \left(u_t - \delta,\ \bar{p} - p_t^{\circ}\right) \qquad \left(c_t - \delta,\ g_t^{\circ}\right),\ \left(c_t + \delta,\ \bar{g} - g_t^{\circ}\right)$$

Why it matters B★ and B priv bracket every result: the forecast is useful only where B fcst stays between them.

How we perturb the price forecast

2 · ERROR MODEL

Forecast used by battery e

$$\hat{\lambda}_{e,t} = \lambda^{\star}_{t} + \underbrace{b\,\bar{\lambda}}_{\text{bias}} + \underbrace{\sigma\,\bar{\lambda}\,z_{e,t}}_{\text{noise}}$$

Common part ξ and battery-specific part η

$$z_{e,t} = \sqrt{\kappa}\,\xi_t + \sqrt{1-\kappa}\,\eta_{e,t} \qquad \text{common + battery-specific}$$

Each part persists over hours (AR(1))

$$x_t = \rho\,x_{t-1} + \sqrt{1-\rho^2}\,w_t, \quad w_t \sim \mathcal{N}(0,1) \qquad \text{for } x = \xi, \eta\text{: unit variance}$$

$\bar{\lambda}$ = mean of $\lambda^{\star}$ over the day, so σ and b are in % of the price level.

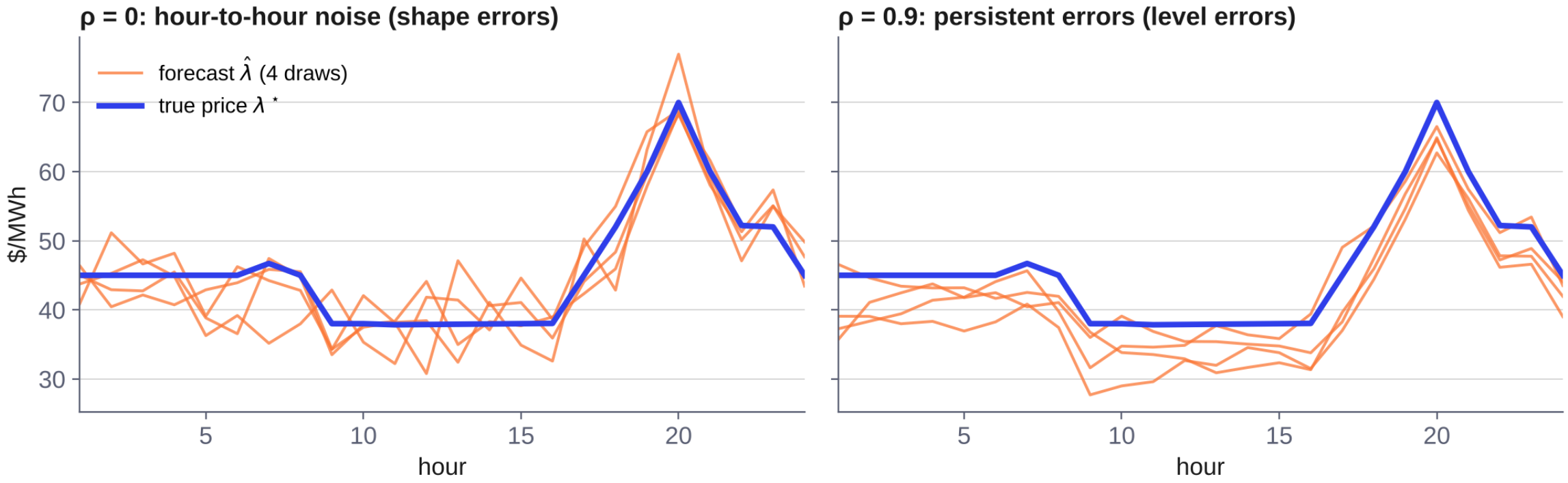
Parameter	What it controls	Values
σ	size of the error	0, 2.5, 5, 10, 20, 40 % of $\bar{\lambda}$
ρ	persistence: shape (0) vs level (0.9) errors	0, 0.5, 0.9
b	systematic bias	−10, 0, +10 % of $\bar{\lambda}$
κ	share common to all batteries	0, 0.5, 1 (several batteries)

Additive and Gaussian: simple, controllable and reproducible. In Stage 3 we replace it by real forecast errors (historical forecast vs outcome).

Why it matters Each parameter isolates one property of a forecast, so we can say which kind of accuracy actually matters.

What the errors look like (σ = 10 % of the mean price)

2 · ERROR MODEL



Size σ: How accurate must a forecast be before bids built on it beat bids built on technical data only?

Persistence ρ: Noise moves single hours (when to charge); persistent errors shift the whole day (how much the energy is worth).

Common share κ: Batteries using the same forecast make the same mistake together and move the price more.

Why it matters Same average error, very different consequences: the experiments separate them.

One Monte Carlo draw, step by step

3 · PIPELINE

1 Benchmark

Solve Baldick's design once: λ★, schedule x★, welfare SW★ (and SW without storage).

2 Forecast

Draw the errors and build $\hat{\lambda}$ for every battery.

3 Bids

Each battery solves its own self-schedule at $\hat{\lambda}$: plan (p°, g°), multipliers, pre-limit rule, bids (E) with quantity steps.

4 Day-ahead clearing

The ISO clears the current design with the true system and these bids: schedule and prices λ CD.

5 Real time

The battery delivers the closest SOC-feasible schedule; other units re-dispatch at offer price + premium ψ; unserved energy at VOLL.

6 Evaluate

Welfare loss, undeliverable energy, price distortion, battery profit.

Why it matters Steps 2 to 6 are repeated N = 500 times per setting: we report the mean, a 95 % interval and the 95th percentile.

What we measure

3 · METRICS

Welfare loss, normalised by what storage is worth in the benchmark

$$\Delta SW = SW^{\star} - SW^{\text{real}} \geq 0, \qquad \text{loss} = \frac{\Delta SW}{SW^{\star} - SW^{\text{no ESS}}} \quad (\% \text{ of storage value})$$

Split into a quantity part and a price part

$$\Delta SW = \underbrace{\Delta SW\left(\mathcal{B}^{\star p}\right)}_{\text{quantity information}} + \underbrace{\Delta SW\left(\mathcal{B}^{\text{fcst}}\right) - \Delta SW\left(\mathcal{B}^{\star p}\right)}_{\text{price forecast}}$$

SW★ ≥ SW real always, because Baldick's design is the welfare optimum over all feasible schedules: the loss is never negative.

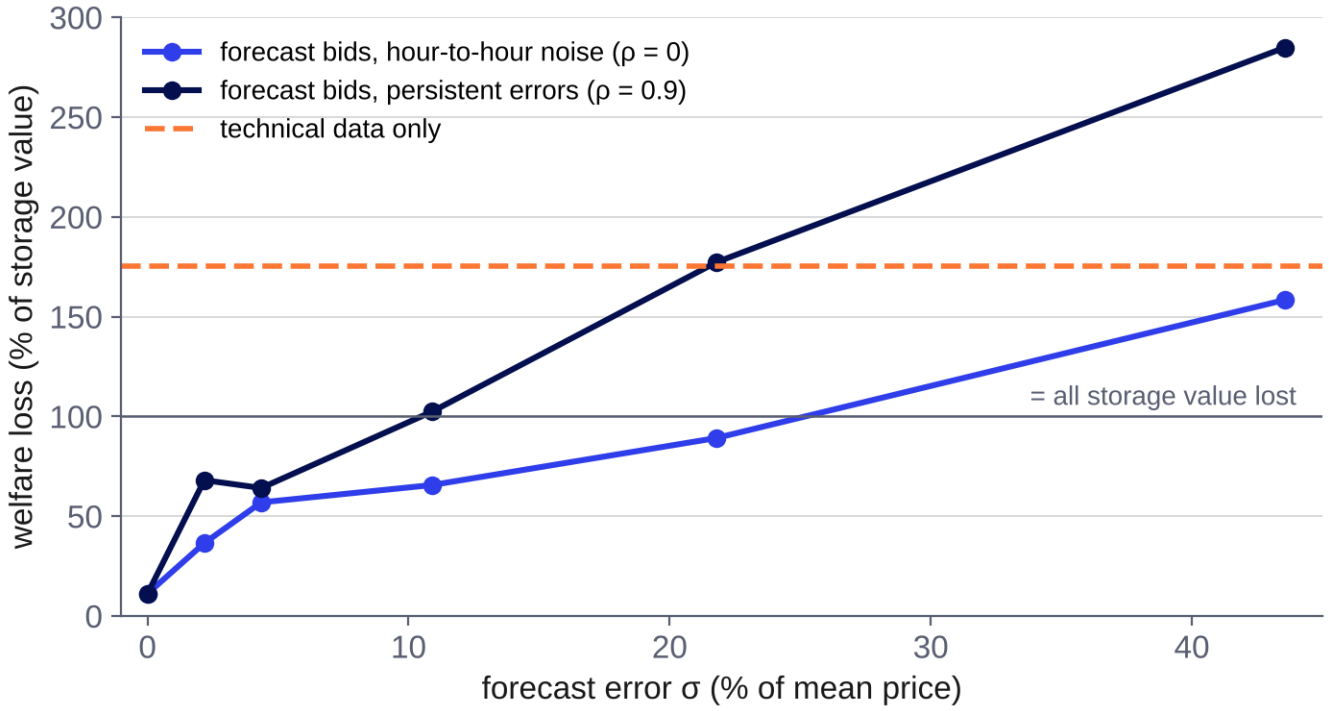
Also reported

- Undeliverable energy (MWh) and SOC-limit violations
- Price distortion: mean |λ CD − λ★|
- Battery profit vs Baldick: who bears the loss
- Value of a forecast: the σ* at which B fcst and B priv lose the same

Why it matters Normalising by the storage value makes results comparable across systems, sizes and days.

Pilot: a 24-hour day, one battery, 300 draws per point

PILOT



Indicative only: synthetic day, battery 100 MW / 400 MWh, ψ = 10 $/MWh, pre-limit rule.

1 **Perfect prices are not enough**

σ = 0 still loses 11 % of the storage value; with Baldick's quantities it is 0. Report quantity and price losses separately.

2 **Persistent errors cost more**

At σ ≥ 10 %, persistent errors lose about twice as much as hour-to-hour noise.

3 **Real-time cost decides the ranking**

Technical-data bids lose 75 % with free real-time adjustment, 175 % with ψ = 10: ψ must be a parameter.

4 **Baldick's 6-hour case is too small**

The battery is larger than the load and sets every price: we use it for validation only.

Why it matters The pilot already shaped the design: quantity rule, real-time premium ψ and the choice of test system.

What we expect, and what the pilot says so far

HYPOTHESES

	Hypothesis	Pilot
H1	True prices and quantities give no loss (Theorem 1).	**confirmed**
H2	The loss grows with σ and comes from the hours where SOC limits bind.	grows with σ; split by hour to do
H3	Persistent errors (ρ = 0.9) cost more than hour-to-hour noise.	**yes, about 2× at σ ≥ 10 %**
H4	The pre-limit rule loses less than the solver's multipliers.	small difference; full grid needed
H5	Errors common to all batteries cost more than independent ones.	Stage 2
H6	A forecast beats technical data only below a threshold σ*, which grows with ψ.	ψ = 10: σ* ≈ 22 % (persistent), > 40 % (noise)

Why it matters Each hypothesis maps to one figure in the thesis, so the experiments have a clear end.

Three stages, from controlled to realistic

PLAN

Stage 1

One battery, 24-hour single-bus day

Full grid: σ (6) × ρ (3) × b (3) × selection rule (2) × ψ (3) = 324 settings × 500 draws. Small LPs: a few hours on a laptop.

Stage 2

Several batteries (5 to 20)

Battery fleet at 10 to 40 % of peak load. Common share κ, fleet size, and how errors interact through the price.

Stage 3

Realistic system and errors

RTS-GMLC area reduced to one bus, many days, higher renewable shares, and real forecast errors instead of Gaussian ones.

Done (Stage 0): theory validated on Baldick's 6-hour cases; pilot of Stage 1 run with the full pipeline.

Why it matters Each stage adds one source of realism, so we can tell which one changes the conclusions.

Decisions I need from you

FOR APPROVAL

1	**Error model**	Additive Gaussian, σ in % of the mean price, persistence (AR(1)) and a common factor. Enough for Stages 1–2, with real errors in Stage 3?
2	**Quantity rule**	Each battery offers its own plan as price steps (± δ). Acceptable, or should it also forecast quantities?
3	**Real-time evaluation**	Closest feasible battery schedule, redispatch premium ψ ∈ {0, 10, 50} $/MWh, unserved energy at 1000 $/MWh.
4	**Main indicator**	Welfare loss as % of the storage value in the benchmark.
5	**Stage 3 system**	RTS-GMLC area (one bus), or another system you prefer? Fleet size?

Why it matters With these five choices fixed, Stage 1 can start immediately: the code for the full pipeline already runs.


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